

MANNAT TRIVEDI

manneducation@gmail.com [9893483451](#) [LinkedIn](#) [GitHub](#) [Leetcode](#)

EDUCATION

B.Tech in Computer Science and Engineering – MANIT Bhopal (2027)
CGPA: 8.64

12th (CBSE Board) – RD Public School, Betul (2023)
91.4%

EXPERIENCE

Software Engineering Intern – Bosch Global Software Technologies (BGSW) (May 2026 – Jul 2026)

- Developed and deployed an internal **Employee Skill Assessment Platform** using **React.js** and **Express.js** for the **TAF Department**, reducing assessment submission time by **40–50%** through bulk uploads and dynamic form generation.
- Implemented secure **Microsoft Entra ID (Azure MSAL)** authentication with **Microsoft Graph API** and **SharePoint** integration, improving enterprise data security, consistency, and centralized data management.
- Engineered scalable backend services for reliable concurrent request handling and successfully deployed the application on **SAP BTP Cloud Foundry**.

PROJECTS

The Binary Critic - Taste-based Media Discovery Platform (*Next.js, Node.js, MongoDB, IndexedDB, NextAuth*)

- Engineered a **Two-Stage Recommendation Engine (Retrieval → Multi-Signal Ranker → MMR)**, cutting feed latency by **67–80% (3.3s to 0.52s)** by pairing a **Metadata similarity algorithm** with batched **\$in aggregations** that reduced DB round-trips from **>50 to 2**.
- Architected a **3-tier caching hierarchy (IndexedDB → Next.js ISR → MongoDB)**, achieving **580,000+ ops/sec client throughput (0.0017ms/op)** and zero memory leaks via a **Hybrid LRU + SWR eviction policy** capped at 50 rails with indexed cursor-based pruning and 7-day TTL sweeps.
- Optimized frontend delivery and search visibility, slashing **LCP by 86% (6.6s to 0.9s)**, **TBT by 67% (120ms to 40ms)**, network payload by **95% (21MB to <1MB)**, and reducing **CLS to 0.01** via layout-stable skeleton streaming, responsive Next/Image WebP delivery, and dynamic **Open Graph structured metadata**.

QueryCraft — Conversational Data Analyst (*AI | Next.js, Node.js, MySQL, llama.cpp, Ollama, Groq, MiniLM, Vectra*)

- Achieved **82.98% accuracy on the BIRD-SQL benchmark (Mini-Dev) Subset** using a **multi-step (Collect–Draft–Critic) reasoning pipeline** for structured NL-to-SQL generation.
- Reduced **schema hallucinations by ~90%** by introducing **local semantic schema retrieval** with **MiniLM embeddings** and a lightweight **Vectra-backed vector index**.
- Built an **on-prem, LLM-agnostic, privacy-first system** supporting **local and cloud inference**, with **read-only enforcement and pre-execution validation** for safe SQL execution.

InvenTrack – Inventory Management System (*Full-Stack | Node.js, Express, MySQL, JWT, bcrypt*)

- Engineered **RBAC with JWT authentication and bcrypt password hashing**, enabling permission-based access.
- Built **product management with image uploads, low-stock alerts, and CSV export**, achieving a **Lighthouse score of 99** with **0ms TBT**.

TECHNICAL SKILLS

- Languages:** C++, C, JavaScript, HTML CSS, Tailwind CSS
- Backend & Web Technologies:** Node.js, Express.js, REST APIs, React.js, Next.js
- Databases:** SQL (MySQL), NoSQL (MongoDB), RDBMS
- CS Core:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks
- Tools & Platforms:** Git, GitHub, Postman, MongoDB Compass, Ollama
- Deployment & Hosting:** Vercel, Railway, Render, Azure

ACHIEVEMENTS

- 250+ DSA Problems:** Solved 250+ problems covering core DSA and problem-solving patterns.
- TidyTrack:** Built and deployed a campus issue tracking Android app for MANIT at tt.manit.ac.in.
- 3rd Place – CodeRush Web Development Contest:** Developed a quiz platform using HTML, CSS, and JS. ([Certificate](#))